date	version	actions
25/05/23	V1	Initialized
25/05/28	V2	Revised (大幅移除原定方式)
25/05/29	V3/v3a	Revised (修改反思部分)

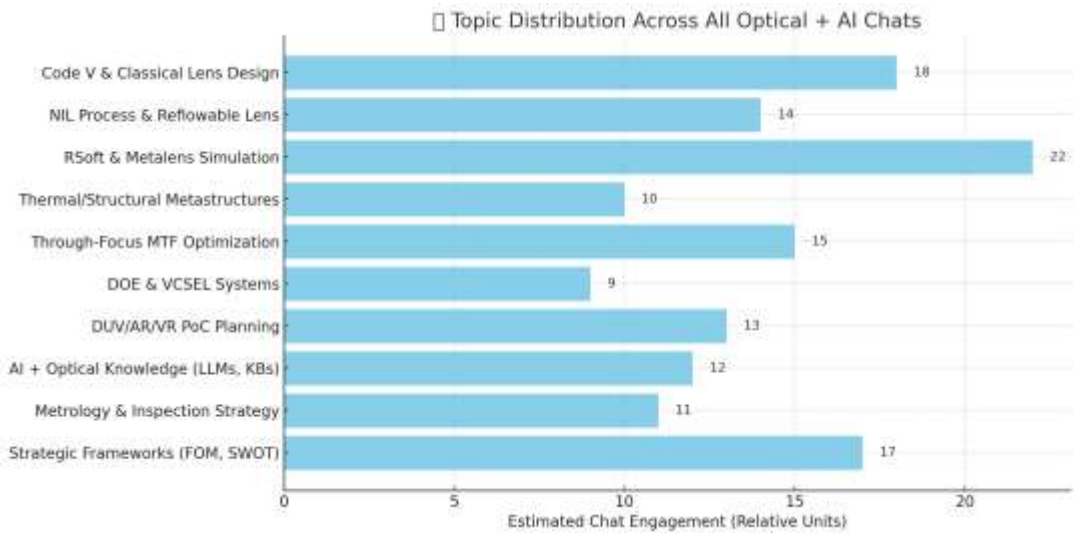
我的光學探索：與生成式AI的奇遇之旅
—重新定義鏡頭設計與分析

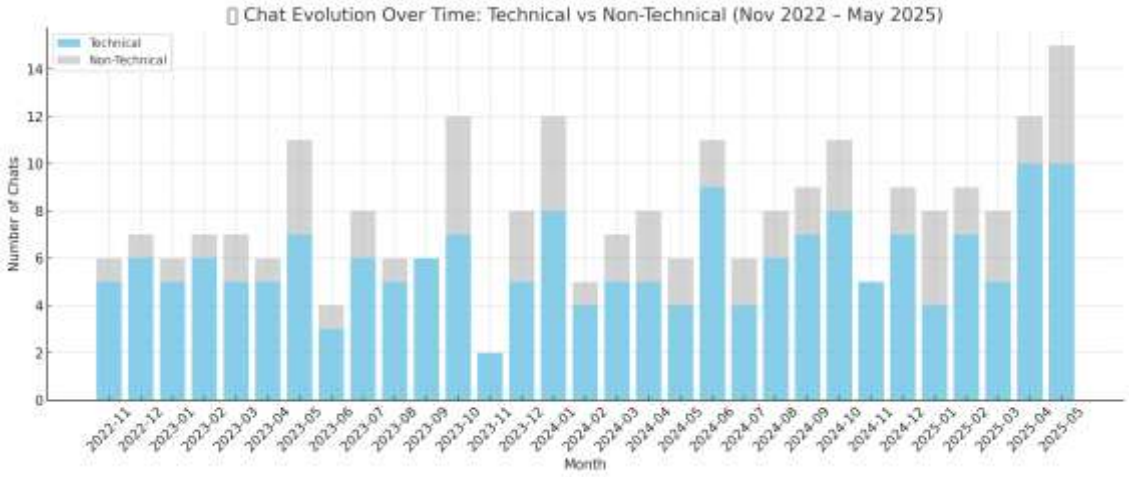
My Optical Exploration with AI:
Redefining Lens Design and Analysis through Generative Intelligence

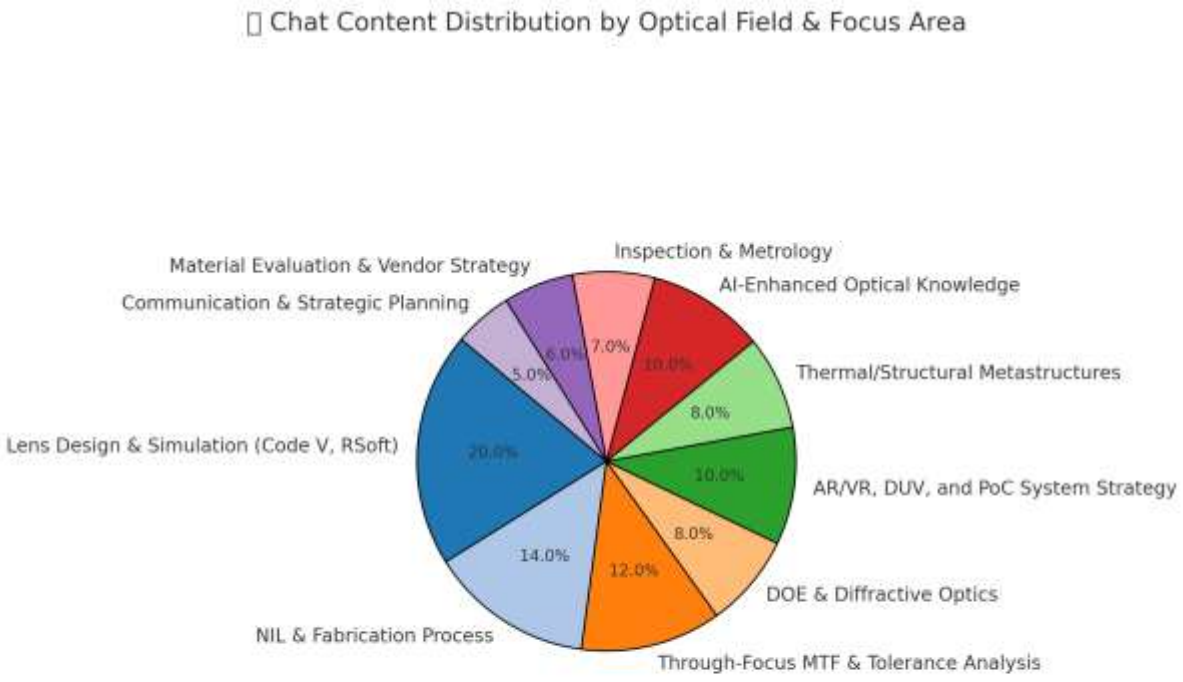
陳志隆 (Jyh-Long Chern)

ChatGPT-我的使用史

- **Over 1,600 total chat messages (turns)**
- engaged in at least **30–35 major sessions** (topics) since the first chat — starting from **late 2022** through **May 2025**.
- Rechecked on May 23



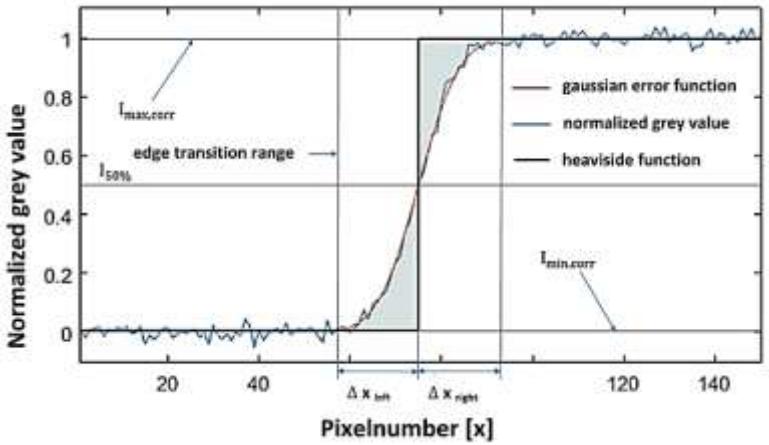




第一次驚豔 – 我約用了一年才覺得不一樣

Derive the fundamental limitation on ETW (edge transit width) in a lens design

(my note in 2023/12)



Edge transition width estimation, method for calculating the image blur. | Download Scientific Diagram

Why this is important?

Performance request – ETW and MTF

- In the case of diffraction-limited optical system, we may ask:
- *What is the limitation on ETW?*
  - Ans: ETW=Diameter of Airy disk
    - ~ 2.44*λ*N
      - λ is the wavelength
      - N is the f/#
- *What is the limitation on MTF?*
  - Ans: MTF=Contrast at cut-off frequency,
    - f_c or f_nyq, f_c=1/(λ*N)
    - If consider Sensor's pixel size, 2.5 um
      - f_c= 1000/2.5~ 200 lp/mm (limited by sensor)
  - MTF curve (ideal case)

$$\mathrm{MTF}(f) = \frac{2}{\pi}\left[\arccos\left(\frac{f}{f_c}\right) - \frac{f}{f_c}\sqrt{1-\left(\frac{f}{f_c}\right)^2}\right]$$

Hence, we have a formula to know what f/# is requested, if ETW is pre-set

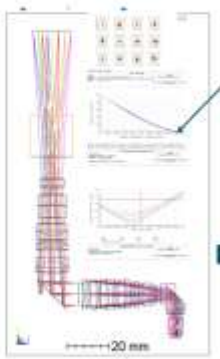
(Here ETW is 0%-100% method)

	Red	Green	Blue	NIR
wavelength (nm)	644	526	470	1150
f/#	ETW (um)	ETW (um)	ETW (um)	ETW (um)
2.8	4.46	3.59	3.21	7.86
4	6.24	5.13	4.59	11.22
5	7.80	6.42	5.75	14.03
6	9.43	7.70	6.88	16.84
7	11.00	8.98	8.03	19.64
8	12.57	10.27	9.17	22.45
9	14.14	11.55	10.32	25.25
10	15.71	12.83	11.47	28.06
11	17.28	14.12	12.61	30.87
12	18.86	15.40	13.76	33.67
13	20.43	16.68	[illegible]	[illegible]
14	[illegible]	17.97	[illegible]	[illegible]
15	[illegible]	[illegible]	[illegible]	[illegible]
16	[illegible]	[illegible]	[illegible]	[illegible]
17	[illegible]	[illegible]	[illegible]	[illegible]
18	26.29	[illegible]	20.64	50.51
19	29.86	24.39	21.79	53.35
20	31.43	25.67	22.94	56.12
21	33.00	26.95	24.08	58.93
22	34.57	28.24	25.23	61.73
23	36.14	29.52	26.38	64.54
24	37.71	30.80	27.52	67.34

Brief:
Given wavelength and target ETW, we must have a f/# which can meet the formula!
*the red-color columns are cited from the specification (see next page); hence, the lens must have the corresponding f/#

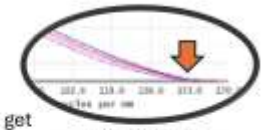
Performance check

Application – 2024/04

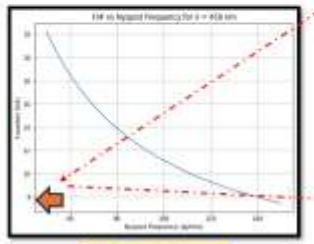


(complex)

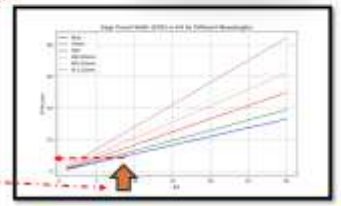
The lens specification (e.g., f/#) is not shown.
The digital in the figure was also not clear.
→ Difficult to identify the f/#
→ Why f/# is important?
Probably ~ 150 lp/mm) (from simplified case)
Because it is with wavelength ~ 450 nm, we can get



(simplified)



→ f/#~ f/7-f/8 (estimated)



So, the best ETW is estimated to be less than 10 um (0-100% ideal edge transition)

2024/4/21 -v2 vendor lens - review and comment 6

case_2024_0_vendor_design review_0419

第一次比較整合式的經驗

陸續約花了兩個禮拜（my work in 2023/0914-26 ）

說明

1. Zemax 裡用的freeform與一般可以見到的freeform (done)
2. 什麼樣的freeform 我們該用?為什麼? (done)
3. 關於Q-type freeform (done)
4. 如何取出Zemax 的 Q-type freeform 的參數 (done)
5. 把取出的Q-type freeform 數據化成超精加工機台可讀的格式 (done)
6. 取出UA3P的面型數據以便進行後續分析 (done)
7. 把UA3P的面型數據用Q-type freeform 做best fitting (done)
8. 把這個best fitted 的Q-type freeform參數倒出 (done)
9. 匯入選定的Zemax 設計檔 (not complete yet)

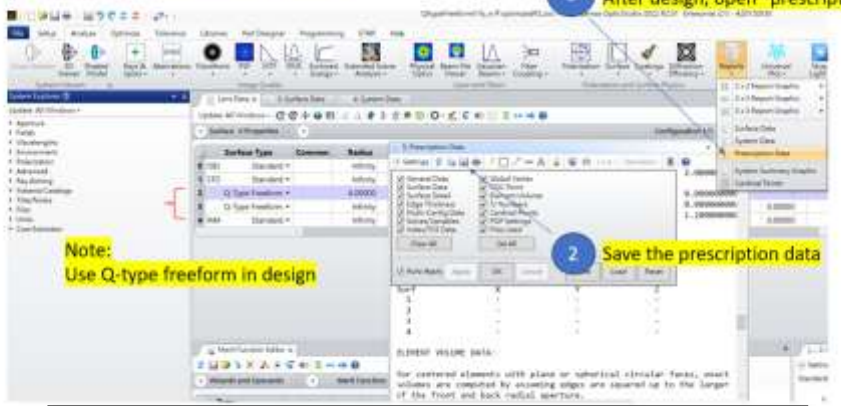
2023/9/26_初稿released (v0)

freeform 精機 - 用Q-type freeform 提升優化模擬與降低製造

Works_myfocus&ff/work_ff surface fitting

Get lens prescription data

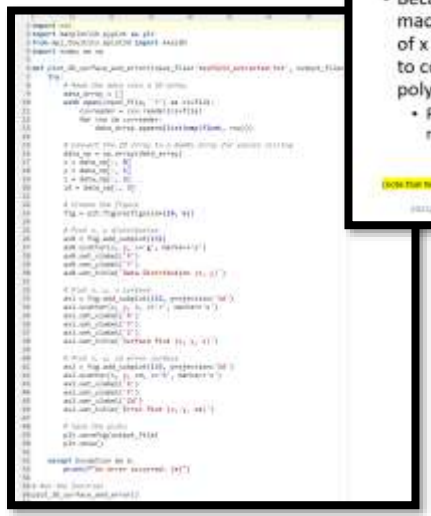
因為過於複雜,所以最好是用自動提取的方式
(所以我們用 prescription data 因為這個資料最全)



Motivation and background

- This work is to convert the Q-type freeform in Zemax to the tooling machine for tooling
- Because our current tooling machine only accepts the form of x and y polynomial, we need to convert our freeform in a polynormal of simply x and y.
  - Please note the tooler recognizes (+ -0.01) as (-0.01)

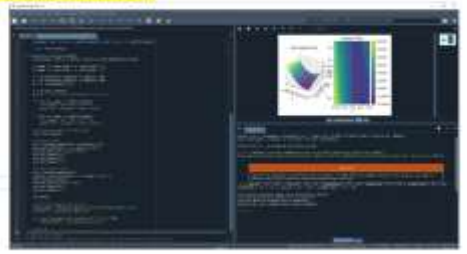
加工機台認得的



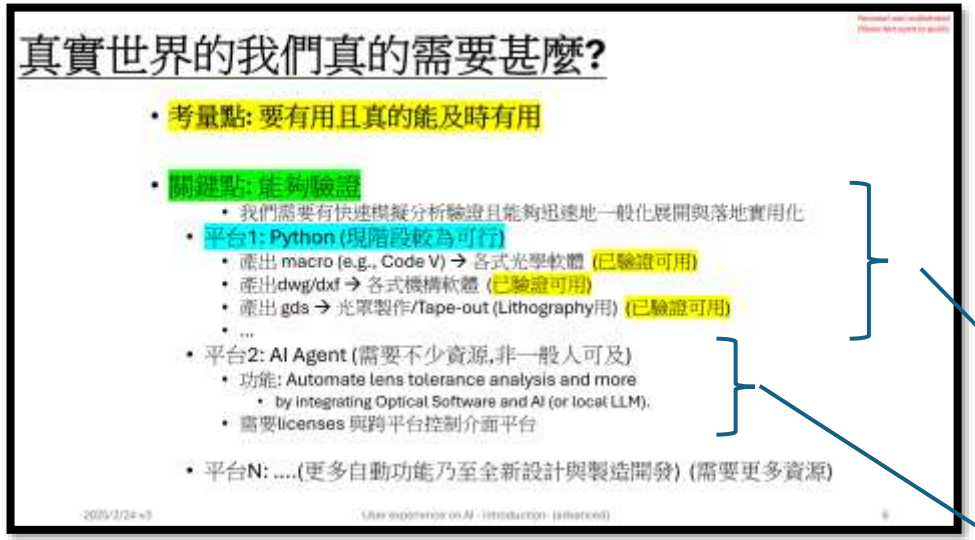
Qtype_generate_surface_code_with_plotsM1F

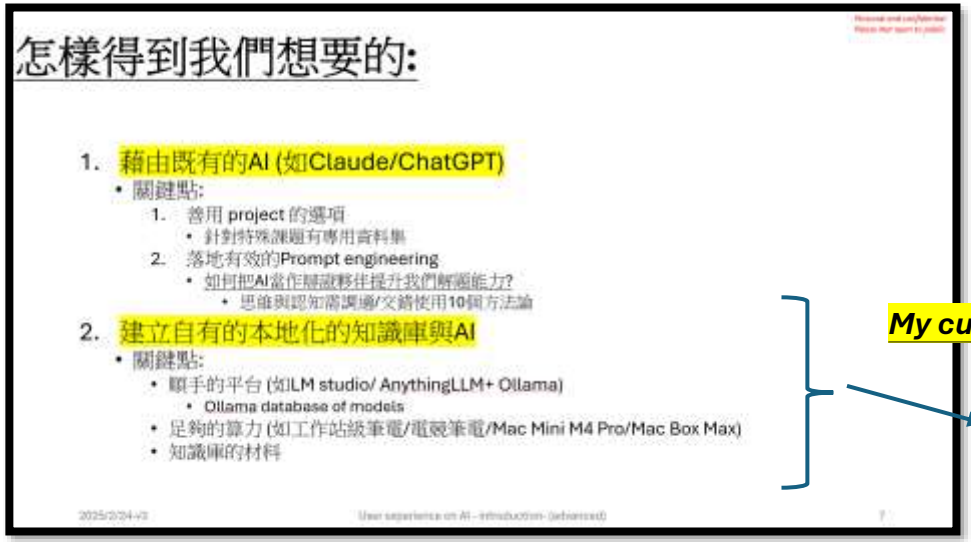
把驗出的參數給在同一個檔案(也一起對應的面型名稱之檔)
流程如下:

1. Read the file and pickup the parameters
   1. Files 如
      surf2_extracted_parameter
2. Plot the surface for cross check
3. print out the formula to console
4. Save to a text file
   1. Parameter_for_tooling



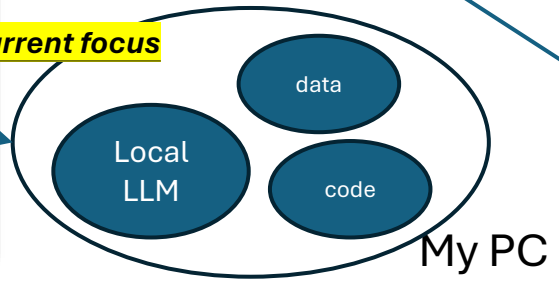
我的之前經驗– 節自25/2/24（v3）





Key breakthrough (to me)	Applications	
產出 macro (e.g., Code V)	各式光學軟體	
產出dwg/dxf	各式機構軟體	Can be stl file for 3D printing
產出 gds	光罩製作/Tape-out (Lithography用)	
產出DOE/Fresnel zone plate/Gabor zone plate	Advanced /自用	
產出metalens	Advanced自用	

My current focus



Opensource works in GitHub

- Lens design
- Blind deconvolution
- Metalens
- 3D image formation – endoscopic
- PINN (physics-informed neural network)

我怎麼做我的設定

問答方式:

Respond in three phases.

Phase 1: As in this Initial phase, simply provides your answer directly fit to my question.

Phase 2: In this second phase, extend my question and respond under the following conditions:

(1) respond as an expert of optics.

(2) responds as an expert of lens design with excellent ability in software, e.g., Code V, LightTools, and Rsoft .

(3) provides further comments, suggestions, and guidance on lens design step by step. List them in a table. Finally,

(4) summarizes your responses and list them in a new table with enough and clear figure of merit, and the corresponding order of magnitude for each item.

Provides the reason why the figure of merit is considered.

As the end of this phase 2,

(1) lists "follow-up" items one by one. List them in a new table.

(2) provides at least "three" different novel, creative, and non-trivial comments with deeper and wider perspectives. Also list them in a new table.

Phase 3: I like you, in this third phase,

(1) re-define the questions based on your database and knowledge which are beyond above two phases 1 and 2,

(2) illustrate your plans/strategy and scopes for these redefined questions,

(3) provide your evaluations on these plans/strategies and scopes. Then,

(4) make comparison with SWOT analysis in a table. Finally,

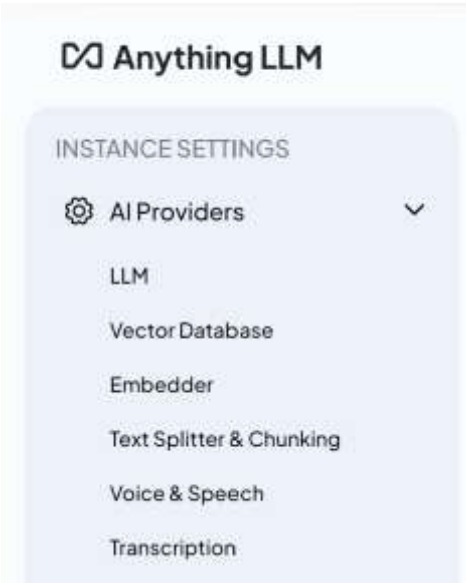
(5) give your answers to these redefined questions and list them in a table.

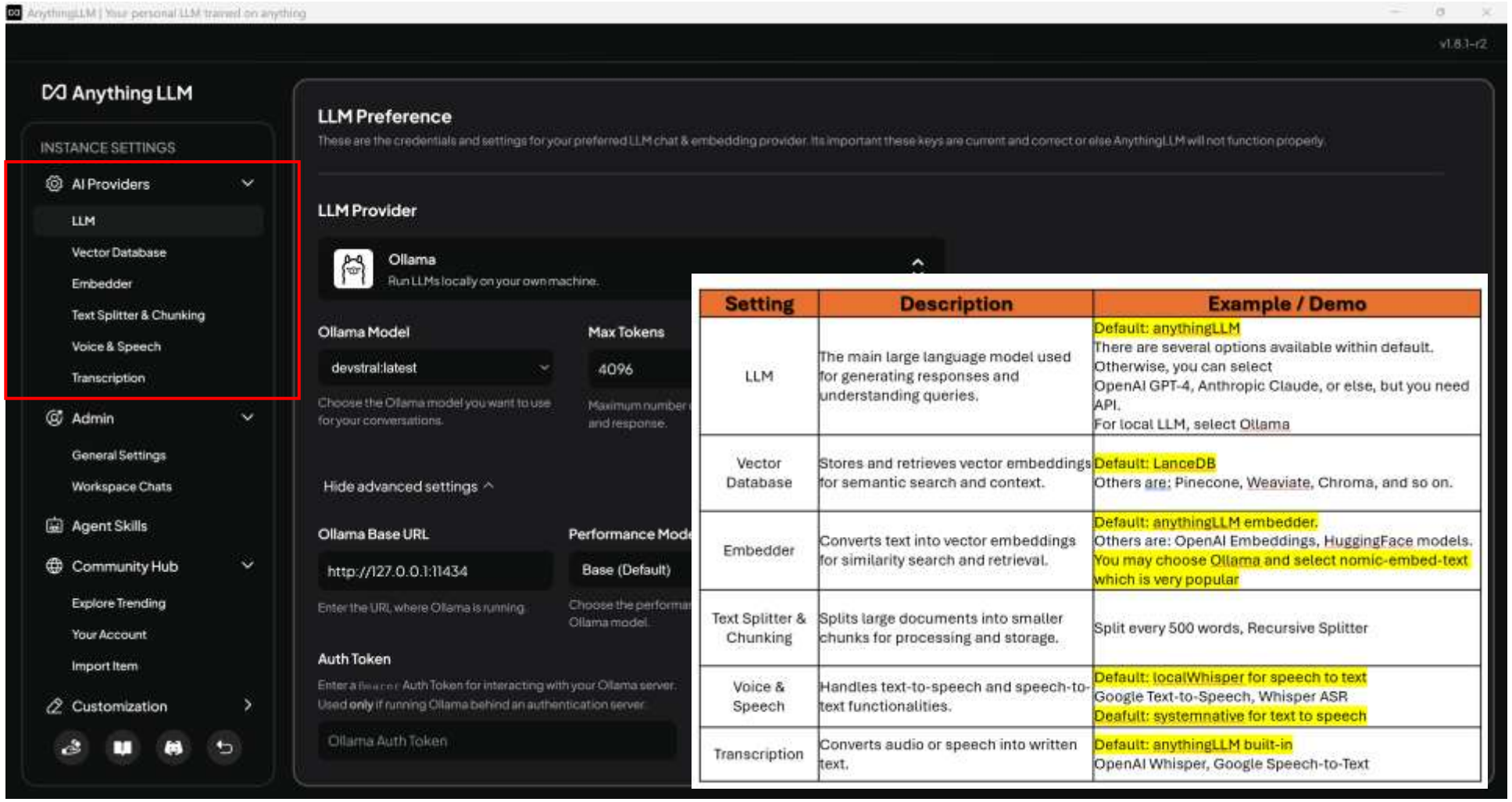
1. 藉由既有的AI (如Claude/ChatGPT)
   - 關鍵點:
     1. 善用 project 的選項
        - 針對特殊課題有專用資料集
     2. 落地有效的Prompt engineering
        - 如何把AI當作辯證夥伴提升我們解題能力?
          - 思維與認知需調適/交錯使用10個方法論

我的焦點:

- My focus is optical technology, particularly lens design, optical analysis with the exploration of novel approaches and solutions.
- The subjects are more on optimization, i.e., to seek for and provide the best solution for the cases or projects.
- Topics covering quantum sensing, illumination and imaging are also the subjects of interest.

GAI是怎麼看我的數據:以Anything LLM為例

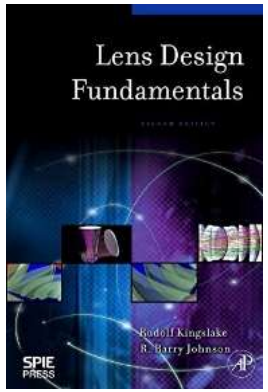




Setting	Description	Example / Demo
LLM	The main large language model used for generating responses and understanding queries.	Default: anythingLLM There are several options available within default. Otherwise, you can select OpenAI GPT-4, Anthropic Claude, or else, but you need API. For local LLM, select Ollama
Vector Database	Stores and retrieves vector embeddings for semantic search and context.	Default: LanceDB Others are: Pinecone, Weaviate, Chroma, and so on.
Embedder	Converts text into vector embeddings for similarity search and retrieval.	Default: anythingLLM embedder. Others are: OpenAI Embeddings, HuggingFace models. You may choose Ollama and select nomic-embed-text which is very popular
Text Splitter & Chunking	Splits large documents into smaller chunks for processing and storage.	Split every 500 words, Recursive Splitter
Voice & Speech	Handles text-to-speech and speech-to-text functionalities.	Default: localWhisper for speech to text Google Text-to-Speech, Whisper ASR Deafult: systemnative for text to speech
Transcription	Converts audio or speech into written text.	Default: anythingLLM built-in OpenAI Whisper, Google Speech-to-Text

光學設計的過往
– 近百年的故事

Slide 3: 光學設計的沿革
(2023/04/25 我有給內部一個序列演講 ,也用這個slide)
那時我用chatgpt 等去解構 lens design fundamental)



Optical design 1937 to 1988 . . . Where to from here?

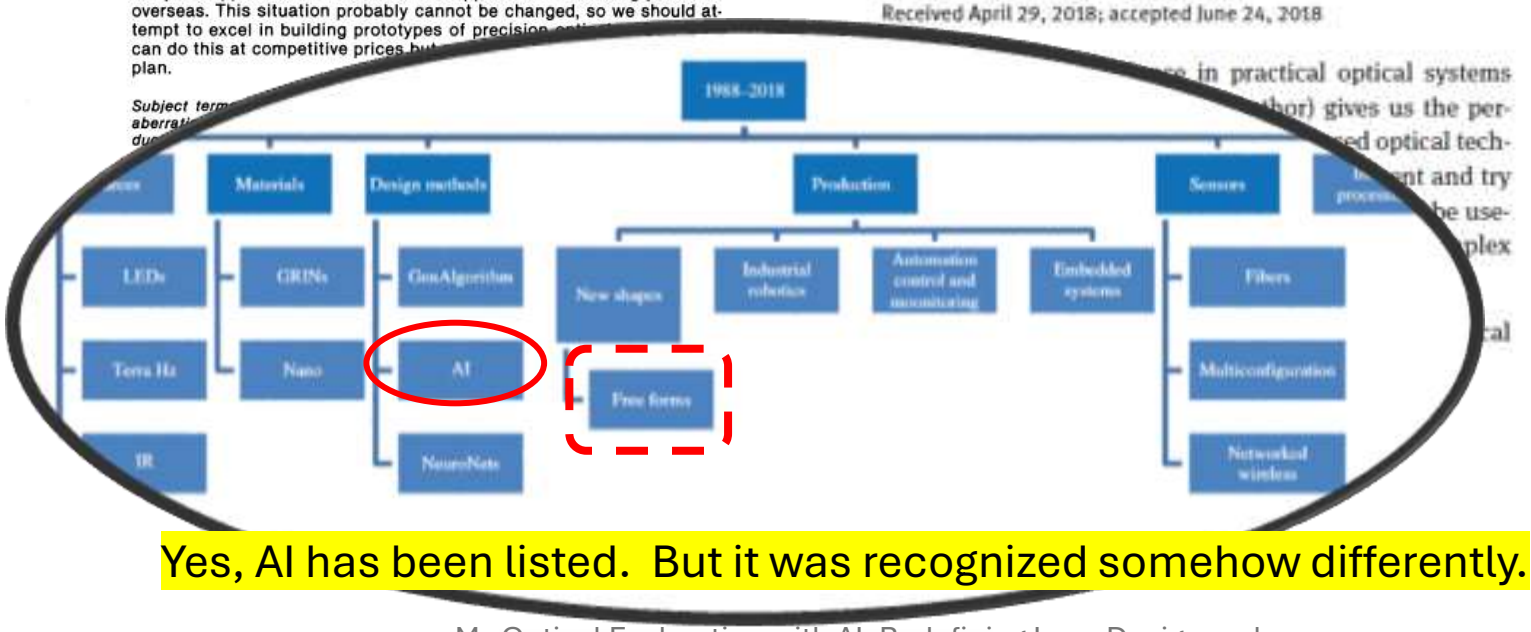
Robert E. Hopkins, FELLOW SPIE
Optizon Corporation
300 East River Road
Rochester, New York 14623

Abstract. By today's standards, lens design and fabrication were primitive during the period from 1937 to 1950, but even today, with all the power of modern computer programs, many of the original techniques are needed. The damped least squares method has been established as the most common method for the optimization process. Several large optical design programs are available commercially. They can handle almost any problem required. They do need many added features, but these can be added to the basic shells. The lens design task today is that of starting with an appropriate understanding of the requirements, a reasonable configuration, and the judgment to be able to make wise trade-offs between conflicting conditions. It is unfortunate that with all our design and manufacturing know-how there are essentially no large production facilities available in the United States. It appears that our skills are being directed primarily to small numbers of units for military or space applications. Most business applications are being produced overseas. This situation probably cannot be changed, so we should attempt to excel in building prototypes of precision
can do this at competitive prices but
plan.

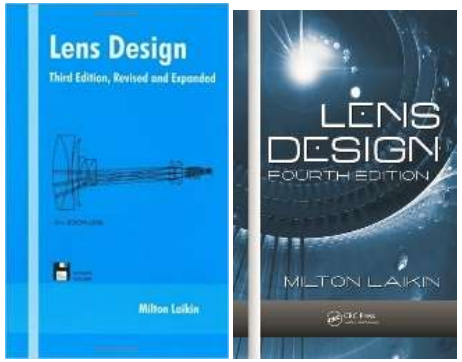
Irina L. Livshits* and Donald C. Dilworth

Trends in optical design from 1988 to 2018… where to from here?

https://doi.org/10.1515/aot-2018-0025
Received April 29, 2018; accepted June 24, 2018



Yes, AI has been listed. But it was recognized somehow differently.



過去的
-光學人是怎麼看?

Slide from my internal talk 23/04/25)

Published in 2018

7 Future Development **44**

7.1 Treatment of patents 44

7.2 Application of deep learning 44

7.3 Common framework of program development 44

7 Future Development

7.1 Treatment of patents

The patent of the lens has a long history. In the old days, lens design was the result of inspiration, trial and error, and patient calculations. A patent was necessary to protect the result of such efforts. Today, a global optimizer creates one solution every minute. Starting from parallel plates, without any intervention from the lens designer, a solution sometimes infringes patents. In this case, constraints need to be imposed to avoid patented designs. It would be ridiculous to claim the novelty of a solution created automatically. It needs to be reconsidered what should be granted as a patent. It is not likely that patent examiners have new ideas about what should be granted patents, so lens designers must provide those ideas.

7.2 Application of deep learning

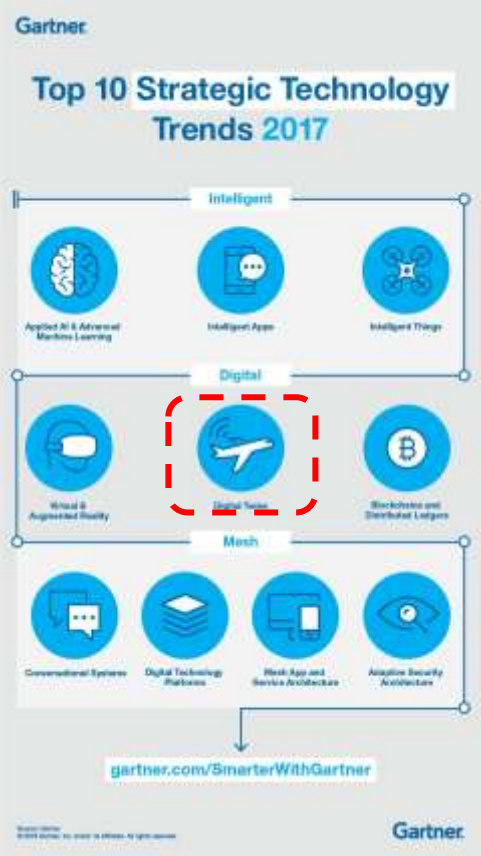
Today, the vast number of solutions is created by the staggering number of PCs in the world. On the other hand, advanced artificial intelligence is expected to extract new insights from the sheer number of data. Lens design could be the subject of this new technology. To analyze solutions and produce new insights, the solutions need to be optimized with a standard and powerful optimizer.

7.3 Common framework of program development

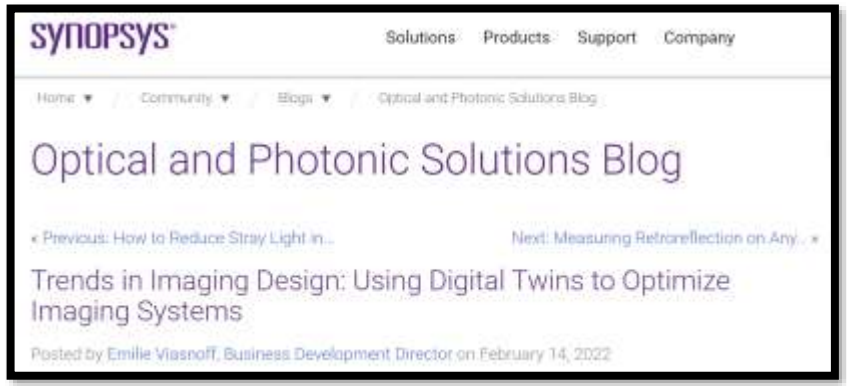
A mathematical library accessible through the internet is useful for programming, and a similar framework would be necessary for the development of lens design codes. It would be valuable to prepare a common data format and common program structure so that many independent programmers can exchange program parts.

甚麼已經被改變了?

Gartner's Top 10 Strategic Technology Trends For 2017



Optical community



https://blogs.synopsys.com/optical-solutions/2022/02/14/trends-in-imaging-design-using-digital-twins-to-optimize-imaging-systems/

https://www.forbes.com/sites/gartnergroup/2016/10/26/gartners-top-10-strategic-technology-trends-for-2017/?sh=47487187186b

https://www.gartner.com/en/articles/gartner-top-10-strategic-technology-trends-for-2023

The world changes as chatgpt comes!

Slide from my internal talk on 23/04/25

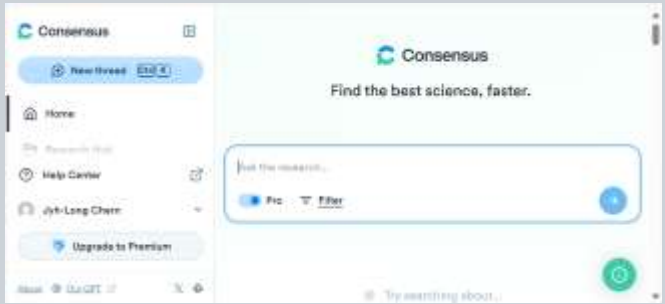
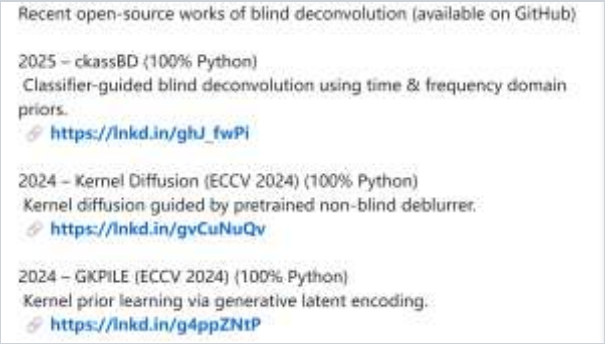
我現在用的GAI

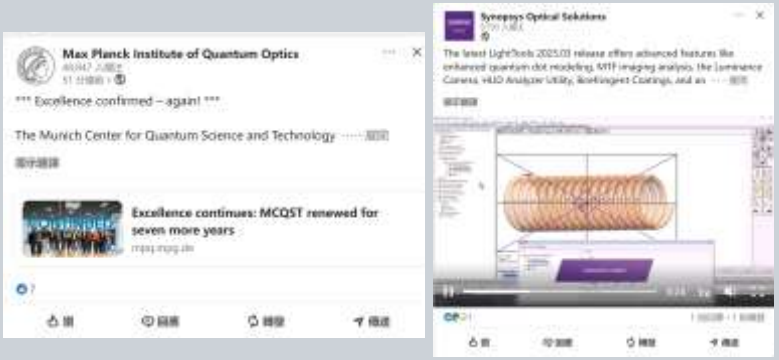
目前我主要用的

Anthropic - Claude (付費版)	Open AI - Chatgpt (付費版)	Google - Gemini (付費版)
Project	Project	Gem
	Deep research	Deep research
273 chats	>1600 chats	
三個GAI答題各有千秋,即便是同一問題.同樣的提示詞答的都不一樣居多.		

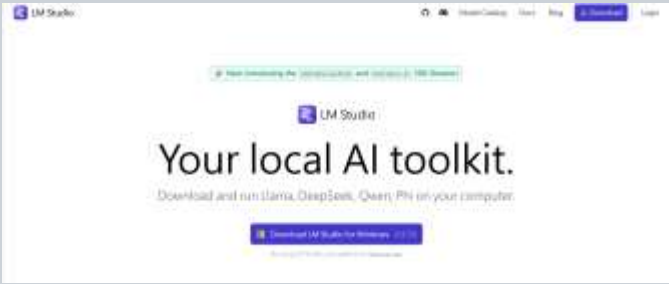
我通常會同一個課題三個都問

我常用的資料庫

Consensus	Github	NotebookLM
		
有用的資料搜尋,資料正確性高	非常多的開源作品可供參考	可以整本書作解讀,可以幫讀語音檔

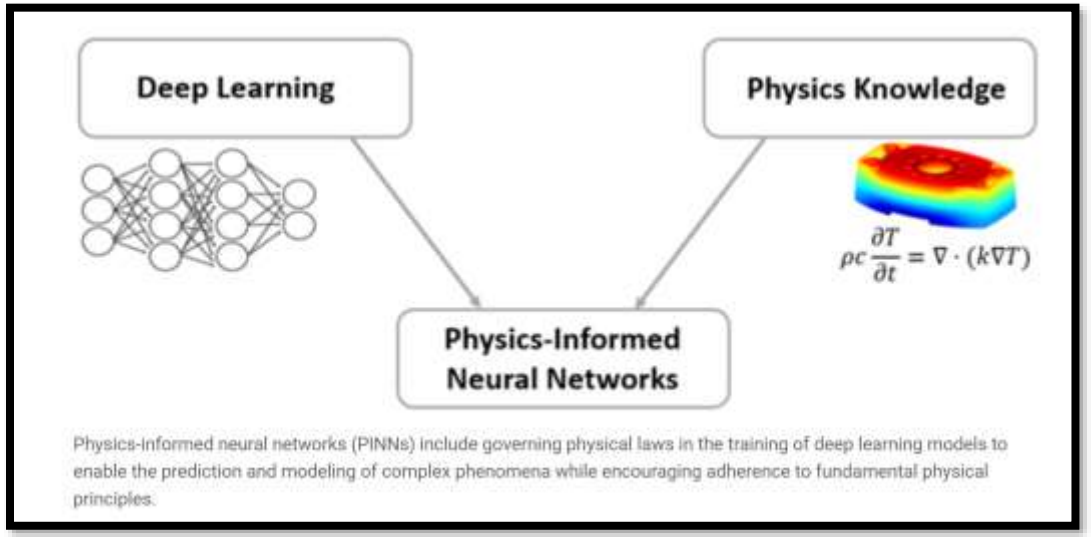
Medium (付費版)	Linkedin	Google 學術搜尋
		
很多很好的文章	與世界各地交友,新知識來源之一	勾選項目 新論文直接通知到信箱

我目前主要用的LLM平台

LM studio	AnythingLLM	Ollama
		
非常好用的界面 (初入門)	可以建專用資料庫	Local LLM 下載 (我通常拿去給anything LLM用)

關於PINN
(physics-informed neural networks)

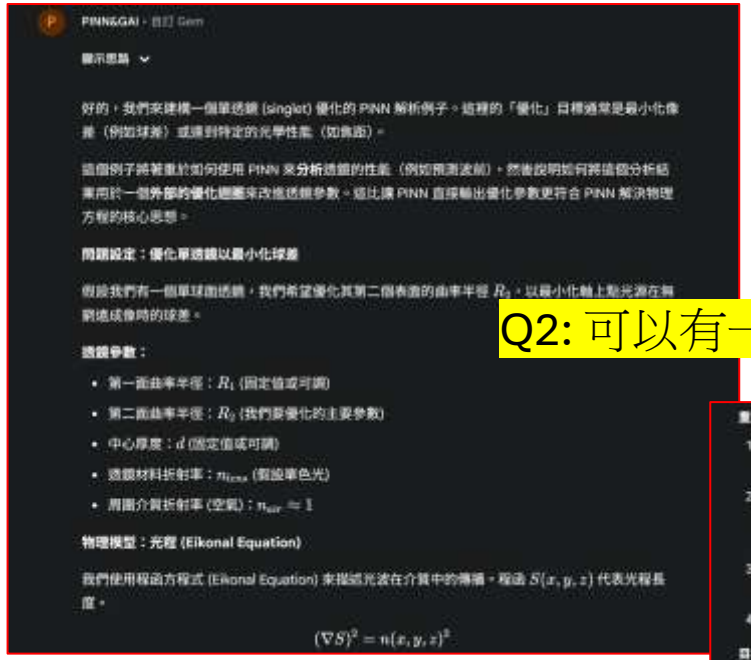
要怎樣才能讓AI比較真的是有物理認知的推理?
---使用 PINN



- What Are Physics-Informed Neural Networks (PINNs)? - MATLAB & Simulink

a significant part of AI-driven optimization and analysis, especially *for ensuring physical plausibility in designs and simulations.*

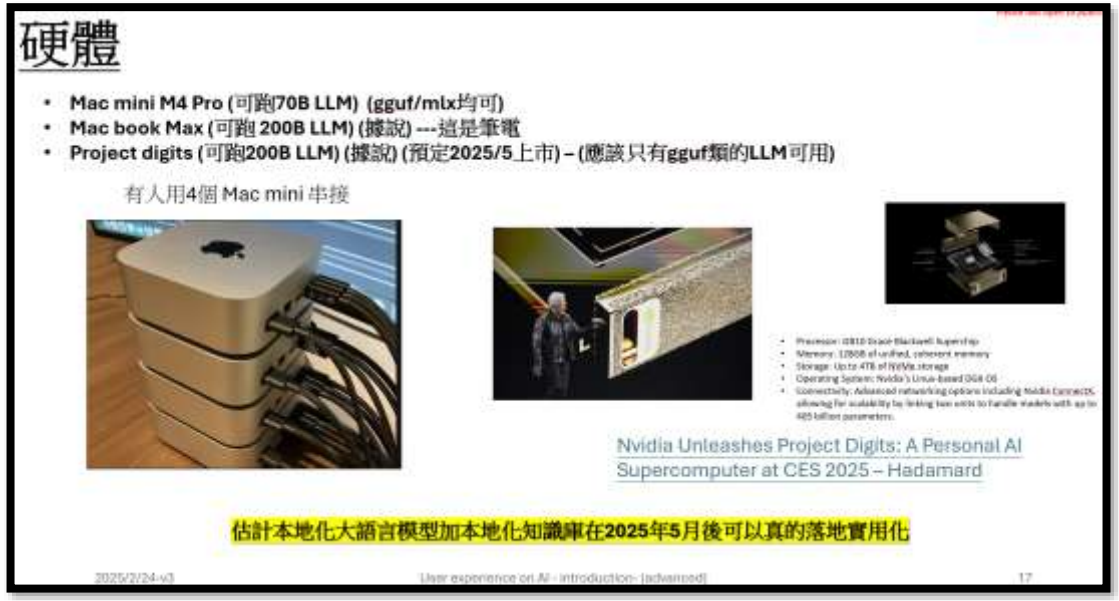
Q1:請給一個單透鏡(singlet)優化的PINN解析例子?



Q2: 可以有一個python code 做演示嗎?

其實也不見得要GitHub, 用GAI也可以產生

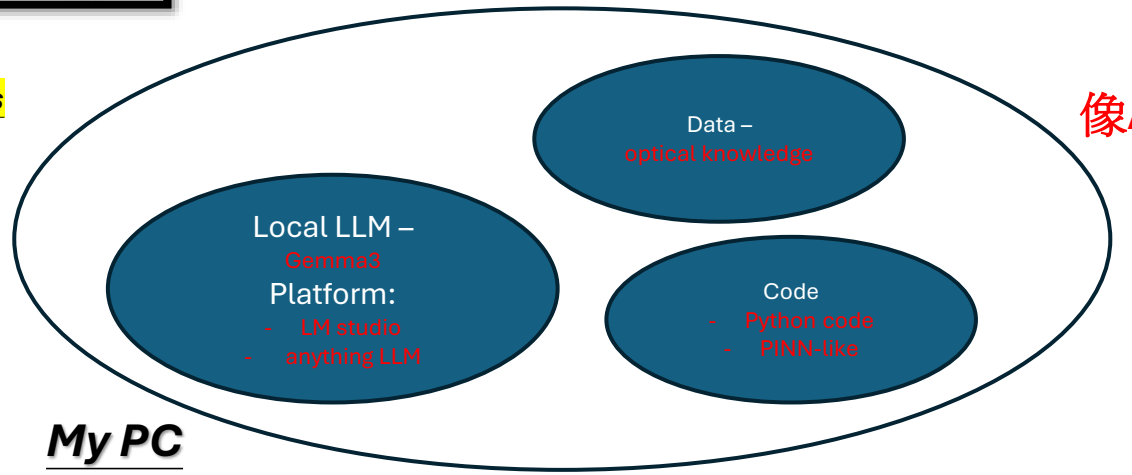
一些關於硬體的事:



DGX Spark具備每秒1 petaflop的AI算力，支援超過千項AI任務，更可擴展成雙機連結達2 petaflops效能，鎖定全球AI開發者與研究機構為主要使用對象，將於年底量產出貨。

與蔡力行Computex同台！黃仁勳揭DGX Spark開發祕辛：第一次就跑起來是奇蹟

My current focus



My PC

像Agent 的架構

反思：

一個反思
作為結尾

這幾天我看到了：

An MIT Exploration of Generative AI • From Novel Chemicals to Opera

From Automation to Augmentation: Redefining Engineering Design and Manufacturing in the Age of NextGen-AI

Md Ferdous Alam[1] Austin Lentsch[1] Nemi Yu[1] Sylvia Barmack[2] Suhin Kim[1] Daron Acemoglu[3] John Hart[1] Simon Johnson[4] Faez Ahmed[1]

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MIT

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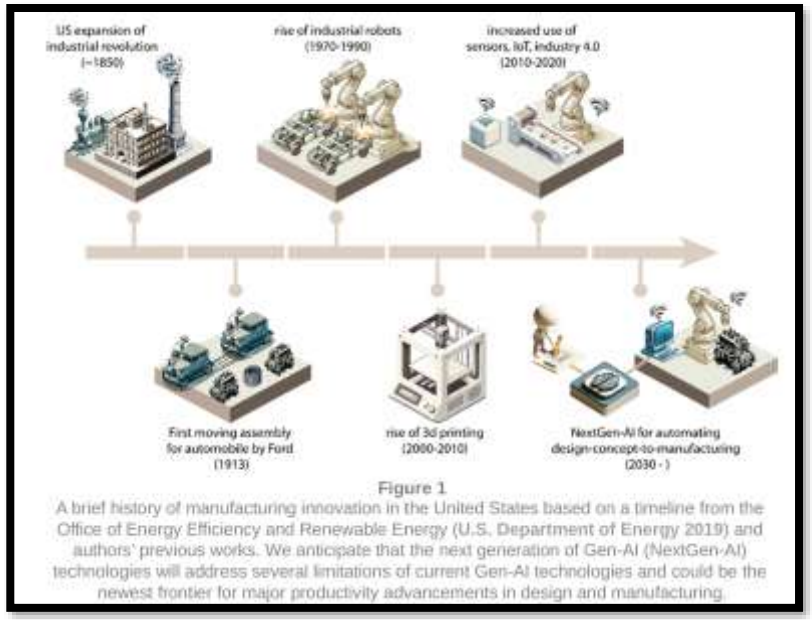


Figure 1
A brief history of manufacturing innovation in the United States based on a timeline from the Office of Energy Efficiency and Renewable Energy (U.S. Department of Energy 2019) and authors' previous works. We anticipate that the next generation of Gen-AI (NextGen-AI) technologies will address several limitations of current Gen-AI technologies and could be the newest frontier for major productivity advancements in design and manufacturing.

光學設計一個演歷
(Lens Design Evolution Timeline)

[1850 - 1970]: Classical Analytical Lens Design
- Manual ray-tracing
- Gaussian optics
- Standard optical formulae (thin-lens equations)

[1970 - 1990]: Software-based Optical Design Automation
- Introduction of software tools (Code V, Zemax, OSLO)
- Computer-aided ray tracing
- Basic automated optimization (sequential ray tracing)

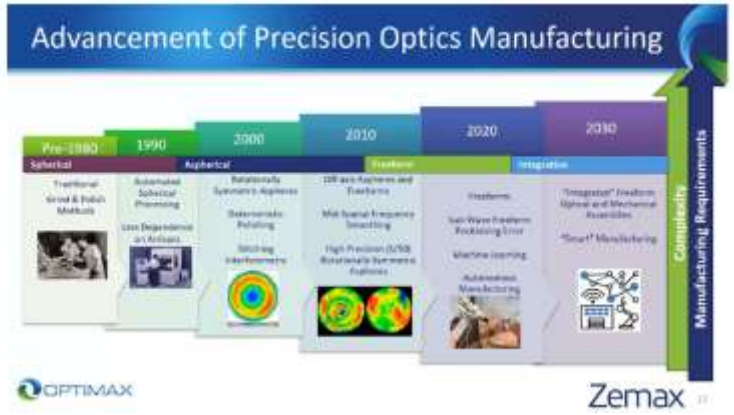
[1990 - 2010]: Advanced Optimization Techniques
- Global optimization algorithms
- Robust tolerancing and sensitivity analysis
  (Design for Manufacturability (DFM))
- Non-sequential ray tracing
- Statistical methods in lens design

[2000 - 2015]: Emergence and Adoption of Freeform Optics
- Non-axisymmetric optical surfaces
- Significant reductions in optical volume and complexity
- Advanced manufacturing and metrology improvements
- Applications in compact imaging, AR/VR, and illumination optics

[2010 - 2020]: Computational Optics and Diffractive Elements
- Rise of Diffractive Optical Elements (DOEs), Metalenses, and Metasurfaces
- Hybrid ray-wave modeling (RSoft, MetaOptic Designer)
- Computational imaging techniques
- Emergence of nanoimprint lithography (NIL) lens fabrication

[2020 - 2030]: Current Era: Generative AI Integration
- AI-driven optimization and tolerancing
- Rapid parameter search and recommendation systems
- Data-driven inverse optical design
- AI-enhanced metrology and manufacturing feedback

[2030 and Beyond]: NextGen-AI and Autonomous Lens Systems
- Fully autonomous optical design and optimization
- Real-time adaptive optical elements
- End-to-end automated design-to-production workflows
- AI-enabled self-optimizing optical systems



- **Aberration theory developments** (1940s-1960s) –
  - The work of Kingslake, Hopkins, and others that bridged classical and computer-aided design
- **Zoom lens revolution** (1960s-1980s) –
  - Mechanical cam design and multi-configuration optimization
- **Aspheric surface manufacturing breakthrough** (1980s-1990s) –
  - Diamond turning and precision molding technologies
- **Plastic optics emergence** (1990s-2000s) –
  - Injection molding for consumer optics
- **Smartphone camera revolution** (2005-2015) –
  - Miniaturization and computational photography integration

我們的**lens design**要能回答

我們的**lens design**
要能回答

- **How can we transition from discrete evolutionary steps to continuous innovation in lens design?**
  - 持續迭代創新
- **What fundamental physical limits remain unaddressed in current design paradigms?**
  - 未墾物理疆域
- **How do we quantify and optimize for "emergence" in optical systems?**
  - 湧現

On Emergence（湧現）

- **Emergence** in optical systems manifests when the collective behavior of optical elements creates new functionalities not inherent in any single component.
- This contrasts with traditional lens design where system performance is largely predictable from individual element contributions.
- 我們的lens design with GAI/Physical AI 要能夠讓我們看到”湧現”.
  - → 目前看起來是可能的

Emergent Property	System Components	Resulting Behavior	Traditional vs Emergent
Computational Imaging	Coded aperture + Algorithm + Detector	Super-resolution beyond diffraction limit	Traditional: Resolution fixed by optics Emergent: Resolution enhanced by system
Metalens Arrays	Subwavelength structures	Anomalous dispersion control	Traditional: Material dispersion fixed Emergent: Engineered dispersion
Adaptive Imaging	Deformable mirror + Wavefront sensor + Control	Real-time aberration correction	Traditional: Static correction Emergent: Dynamic optimization
Light Field Cameras	Microlens array + Sensor + Processing	Post-capture refocusing	Traditional: Fixed focal plane Emergent: Computational focus
Quantum Imaging	Entangled photons + Coincidence detection	Ghost imaging, sub-shot noise	Traditional: Classical limits Emergent: Quantum advantages

On Continuous Innovation (持續迭代創新）－（湧創）

- **Continuous Innovation in Lens Design** means shifting from periodic breakthroughs (like the jump from ray tracing to optimization software) to a constant stream of incremental improvements through real-time feedback loops between design, manufacturing, and application.
- Traditional lens design has progressed through discrete revolutionary steps - each lasting 20-30 years.
- The future demands a fluid, constantly evolving design ecosystem.
- 我們的lens design with GAI/Physical AI 要能夠讓我們看到湧創.
  - → 目前看起來是可能的

Aspect	Discrete Evolution (Current)	Continuous Innovation (Proposed)	Transformation Required
Design Cycle	6-24 months per design	Real-time adaptation (hours/days)	Automated design loops
Feedback Integration	Post-production analysis	In-situ performance monitoring	Embedded sensors + IoT
Knowledge Transfer	Conference papers, patents	Shared AI models, cloud databases	Open design platforms
Optimization	Fixed merit functions	Dynamic merit function evolution	Adaptive algorithms
Manufacturing Link	Sequential handoff	Concurrent design-manufacture	Digital twin integration

On Unaddressed Fundamental Physical Limits (未墾物理疆域）－（湧視）

- **Unaddressed Fundamental Physical Limits** refers to physical phenomena and constraints that current lens design paradigms either ignore, work around, or haven't yet exploited - essentially, the physics we know exists but haven't integrated into optical design methodology.
- The list here may be too under-estimated since indeed, there exist pioneering works already
  - Particularly on meta-devices for thermal-related applications
    - E.g, Thermal Cloaks, Rotators, and Concentrators
- 我們的lens design with GAI/Physical AI 要能夠讓我們看到”湧視.“
  - → 目前看起來是可能的

Physical Phenomenon	Current Treatment	Untapped Potential	Why Unaddressed
Quantum Noise	Seen as limitation	Information source	Lack of quantum optics integration
Thermodynamic Limits	Ignored in design	Energy optimization possible	No thermal-optical co-design
Vacuum Fluctuations	Not considered	Casimir-based effects	Too exotic for classical design
Nonlinear Dispersion	Avoided or minimized	Soliton propagation benefits	Complex mathematics
Spin-Orbit Coupling	Unknown in lens design	Geometric phase control	Recent discovery (2015+)
Topology	Not exploited	Protected optical states	Requires new materials
Time Reversal	Static designs only	Phase conjugation benefits	Dynamic complexity